Unit 6, Lesson 10

Using Algorithms with Partial Products: 2 Two-Digit Numbers

Standards

Addressing	**4.NBT.B.5**
Building Toward	**4.NBT.B.5**

Instructional Routines

- MLR1 Stronger and Clearer Each Time
- Notice and Wonder

Goals

- Interpret (orally and in writing) an algorithm that uses partial products to multiply 2 two-digit numbers.
- Use an algorithm that uses partial products to multiply 2 two-digit numbers.

Student Facing Learning Goals

Let's try to multiply two-digit numbers with an algorithm that uses partial products.

Lesson Purpose

The purpose of this lesson is for students to use partial products in an algorithm to multiply 2 two-digit numbers.

Narrative

In the previous lesson, students learned to record the partial products from multiplying vertically, using an algorithm. They made connections between the new notation to the structure of the rectangular diagram on which they used earlier. In this lesson, students apply that work to 2 two-digit factors.

Prior to this lesson, students have not needed to compose units when finding the sum of partial products. In the *Warm-up*, students are invited to compare the recording of composed units in both an addition problem and in the sum of the partial products in a multiplication problem.

Access for Students with Disabilities

- Representation

Access for English Learners

- MLR8

Lesson Timeline

Warm-up	10 min
Activity 1	25 min
Activity 2	10 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

How did understanding the *Cool-down* of the lesson before you started teaching today help you synthesize that learning?

Warm-up

10 min

Notice and Wonder: Ways to Keep Track

Standards

Building Toward **4.NBT.B.5**

Instructional Routines

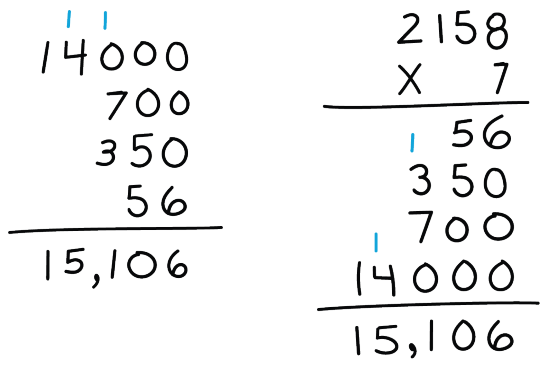
- Notice and Wonder

The purpose of this *Warm-up* is for students to compare ways to keep track of composing units in addition problems and in multiplication problems in which the partial products have a newly composed unit. While students may notice and wonder many things about these ways of recording, the important discussion point is to invite students to consider whatever way they currently record composing a new unit when they add, and then to use it when they multiply if there's a need to compose a new unit when they add up the partial products.

This prompt gives students opportunities to look for and make use of structure (MP7) as they attend to ways to record newly composed units when they add partial products.

Student Task Statement

What do you notice? What do you wonder?



Student Response

Students may notice:

- The numbers that are added up are the same in each problem but their order is different.
- Some columns have little 1s.
- In both problems, there is a small 1 written in the hundreds and thousands place.

Students may wonder:

- Why is there a 1 in the hundreds (or thousands) column?
- What does the 1 in the hundreds (or thousands) column represent?
- Does it matter how you record the work?

Launch

- Groups of 2
- Display the image.
- "What do you notice? What do you wonder?"
- 1 minute: quiet think time

Activity

- "Discuss your thinking with your partner."
- 1 minute: partner discussion
- Share and record responses.

Activity Synthesis

- "In both examples, the same numbers are added up. How would you explain what happens when adding the tens and hundreds?" (Add 5 tens and 5 tens to get 10 tens, which is 1 hundred or 100. Show that 100 with 0 in the tens place and either record a 1 at the top. To add the hundreds, 300 +700+100=1100, so record 1 in the hundreds place and then show that we also have 1000 to add.)
- "How does place value help us understand what the additional 1s represent?" (If the 1 is in the hundreds column, it represents 100. If it is in the thousands column, it represents 1000.)
- "When you add up partial products, keep track of the composed units in a way that makes sense to you."

Activity 1

25 min

Partial Products, Recorded

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- MLR1 Stronger and Clearer Each Time

In this activity, students analyze multiplication involving 2 two-digit factors. Students make sense of work where the partial products are recorded, but there is no indication of where each product came from. After making sense of work, students use the same strategy of finding and keeping track of partial products to evaluate 31×15.

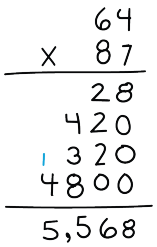
This activity uses *MLR1 Stronger and Clearer Each Time*. Advances: reading, writing

Access for Students with Disabilities

Representation: Internalize Comprehension. Provide students with a partially completed version of the itemized steps as described in the *Activity Synthesis*. For example, offer students a page that shows four copies of 64×87. Each equation could have one of the partial products written in its corresponding color underneath. Invite students to use a green, yellow, red, and purple colored pencil to highlight the digits involved in each step and to write the corresponding multiplication expression next to each partial product.
Supports accessibility for: Visual Spatial Processing, Organization, Attention

Student Task Statement

1. Tyler uses an algorithm to find the value of 64×87.



   Describe each step in Tyler's method. How do you think he arrived at the last five numbers? Record your thinking.

2. Use Tyler's method to find the value of 31×15. Then draw a diagram to check your answer.

Student Response

1. Sample response:
   - 28 is the product of 7 in 87 and 4 in 64.
   - 420 is the product of 7 in 87 and 60 in 64.
   - 320 is the product of 80 in 87 and 4 in 64.

Launch

- Groups of 2
- "Find the value of 64×87 and share your reasoning with a partner."

Activity

- "Take a few minutes to make sense of Tyler's calculation. Be prepared to explain your thinking."
- 3–4 minutes: independent work time on the first problem

MLR1 Stronger and Clearer Each Time

- "Share your analysis of Tyler's calculation with a partner. Take turns being the speaker and the listener. If you are the speaker, share your ideas and writing so far. If you are the listener, ask questions and give feedback to help your partner improve their work."
- 2 minutes: structured partner discussion
- Consider displaying these prompts to support students' conversations:
  - "Can you explain how multiplying ____ and ____ gives ____?"

- 4,800 is the product of 80 in 87 and 60 in 64.
- 5,568 is the sum of the four partial products above it.

2. Sample responses:

$$\begin{array}{r} 31 \\ \times\ 15 \\ \hline 5 \\ 150 \\ 10 \\ +\ 300 \\ \hline 465 \end{array}$$

	30	1
10	$10 \times 30 = 300$	$10 \times 1 = 10$
5	$5 \times 30 = 150$	$5 \times 1 = 5$

- "Can you use the phrase 'partial products' in your explanation?"

- Repeat with 1–2 different partners.
- "Revise your initial draft based on the feedback you got from your partners."
- 2–3 minutes: independent work time
- "Now try using Tyler's method to complete the last problem and use a diagram to check your work."
- 5 minutes: independent work time on the last problem

Activity Synthesis

- To clarify the steps in Tyler's notation, consider itemizing each step and using color coding, as shown here:

$$\begin{array}{r} 64 \\ \times\ 87 \\ \hline 28 \end{array} \quad 7 \times 4$$

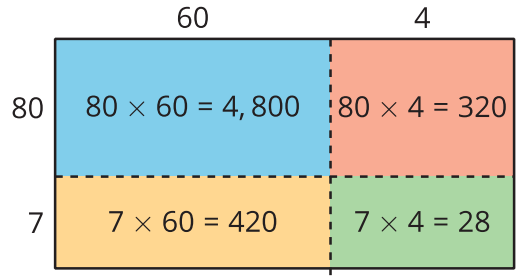
$$\begin{array}{r} 64 \\ \times\ 87 \\ \hline 420 \end{array} \quad 7 \times 60$$

$$\begin{array}{r} 64 \\ \times\ 87 \\ \hline 320 \end{array} \quad 80 \times 4$$

$$\begin{array}{r} 64 \\ \times\ 87 \\ \hline 4,800 \end{array} \quad 80 \times 60$$

$$\begin{array}{r} 64 \\ \times\ 87 \\ \hline 28 \quad 7 \times 4 \\ 420 \quad 7 \times 60 \\ 320 \quad 80 \times 4 \\ +\ 4,800 \quad 80 \times 60 \\ \hline 5,568 \end{array}$$

- Also consider displaying a corresponding diagram for 64×87 and inviting students to make connections to an algorithm.



- "Why did Tyler record a small 1 in the thousands place?" (It shows that we composed a new thousand when adding the values in the hundreds place because 15 hundreds is composed of 1 thousand and 5 hundreds.)

Activity 2

10 min

Han's Multiplication Mishap

Standards

Addressing **4.NBT.B.5**

When using an algorithm that uses partial products, students may be inclined to pay attention only to single digits in each number and pay little attention to the value of the digits. For example, 32×19 might sound like "9 times 2 is 18 and 9 times 3 is 27." In this activity, students analyze this error (MP3) and also look at the commutativity of multiplication when finding partial products.

To help students see that the place value of the digits impacts each partial product, consider displaying a diagram that shows partial products alongside the vertical notation.

Access for English Language Learners

MLR8 Discussion Supports. Synthesis: Display sentence frames to support whole-class discussion: "____ and ____ are the same/alike because . . ." and "Why did you . . .?"
Advances: Speaking, Conversing

Student Task Statement

1. Decide with your partner who will find each product. Show your thinking. Organize it so it can be followed by others.

$$\begin{array}{r} 19 \\ \times\ 32 \\ \hline \end{array} \qquad \begin{array}{r} 32 \\ \times\ 19 \\ \hline \end{array}$$

2. Here is Han's computation of 81×47.

Launch

- Groups of 2

Activity

- "Work with your partner on the first problem. Each partner should find the value of one product."
- 3–4 minutes: partner work time on the first problem
- Pause for a discussion. Select a group whose

$$\begin{array}{r} 81 \\ \times\ 47 \\ \hline 7 \\ 56 \\ 40 \\ 320 \\ \hline 423 \end{array}$$

a. What error or errors did Han make?

b. Show the correct computation for finding the value of 81×47.

$$\begin{array}{r} 8\ 1 \\ \times\ 4\ 7 \\ \hline \end{array}$$

Student Response

1.

$$\begin{array}{r} 1\ 9 \\ \times\ 3\ 2 \\ \hline 1\ 8 \\ 2\ 0 \\ 2\ 7\ 0 \\ +\ 3\ 0\ 0 \\ \hline 6\ 0\ 8 \end{array} \qquad \begin{array}{r} 3\ 2 \\ \times\ 1\ 9 \\ \hline 1\ 8 \\ 2\ 7\ 0 \\ 2\ 0 \\ +\ 3\ 0\ 0 \\ \hline 6\ 0\ 8 \end{array}$$

2. a. Han found 7×5 instead of 7×50. He also found 40×5 instead of 40×50. The second and fourth partial products should be 350 and 2,000, not 35 and 200.

b.

$$\begin{array}{r} 8\ 1 \\ \times\ 4\ 7 \\ \hline 7 \\ 5\ 6\ 0 \\ 4\ 0 \\ +\ 3,\ 2\ 0\ 0 \\ \hline 3,\ 8\ 0\ 7 \end{array}$$

calculations yield the same product to display their work.

- "Should the calculations show the same result? Why or why not?" (Yes, because the same two numbers are being multiplied.)
- "How are the two calculations different?" (The partial products are written in different orders.)
- "Does it matter which number is listed first and which is listed second?" (No)
- "Complete the last problem independently."
- 3–4 minutes: independent work time
- Monitor for students who record the partial products and the newly composed hundred in different ways to correct Han's mishap.

Activity Synthesis

- "In the *Lesson Synthesis*, let's analyze different ways to keep track of partial products and share how we can learn from Han's miscalculation".

Lesson Synthesis

"Today we used partial products in an algorithm to multiply pairs of two-digit numbers. We wrote expressions and equations to help us find and keep track of the partial products as we were multiplying. Sometimes when adding partial products we also needed to keep track of a newly composed hundred."

"Let's look at strategies to correct Han's mishap with solving 81×47."

Display one example of student work side by side with Han's miscalculation.

"Before we analyze this sample closely, can you tell whether the answer is correct or seems more reasonable than Han's work? Why or why not?" (Han's answer was too low because $80 \times 40 = 3{,}200$, so that means the correct answer is

greater than 3,200.)

Select students to explain each step in the computation and any questions the sample raises. Consider recording the factors for each partial product and drawing a diagram to support students in making sense of the products of the decomposed factors in the vertical calculation.

Display a different calculation for 81×47 side by side with the work that was discussed.

"How does this work keep track of partial products and their sum in a different way?"

Reiterate the importance of paying attention to the place value of each digit being multiplied as we find and record the partial products.

Suggested Centers

- Five in a Row: Multiplication (3–5), Stage 3: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 3: Multiply within 100 (Supporting)

Cool-down

5 min

Choose Your Own Strategy

Standards

Addressing **4.NBT.B.5**

Student Task Statement

Find the value of 15×43. Show your reasoning.

Student Response

645. Sample responses:

$15 \times 40 + 15 \times 3 = 600 + 45 = 645$

$$\begin{array}{rrrr} & & 1 & 5 \\ \times & & 4 & 3 \\ \hline & & 1 & 5 \\ & & 3 & 0 \\ & 2 & 0 & 0 \\ + & 4 & 0 & 0 \\ \hline & 6 & 4 & 5 \end{array}$$

Responding to Student Thinking

Students multiply the digits without accounting for their place value (for example, treating 4 and 1 as ones, rather than 4 tens and 1 ten), resulting in inaccurate partial products.

Next Day Supports
Before the *Warm-up*, have students discuss in small groups the partial products for this problem and this specific error.